
Turn in 8

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MA 542 Section A1

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Let $\mathcal{A} \subseteq \mathbb{C}$ denote the set of algebraic integers. We trivially have $0, 1 \in \mathcal{A}$ since $x, x - 1 \in \mathbb{Z}[x]$. Let $\alpha \in \mathcal{A}$. It follows that α is the root of some monic $f(x) = x^n + \cdots + a_0$ in $\mathbb{Z}[x]$. Now consider the polynomial

$$g(x) := f(-x) = \sum_{i=0}^n a_i(-x)^i = \sum_{i=0}^n (-1)^i a_i x^i \in \mathbb{Z}[x].$$

If n is even, then $g(x)$ is monic, and moreover

$$g(-\alpha) = f(-(-\alpha)) = f(\alpha) = 0.$$

If n is odd, then the polynomial $-g(x)$ is monic, and we have

$$-g(-\alpha) = -f(-(\alpha)) = -0 = 0.$$

Moreover, since $g \in \mathbb{Z}[x]$, by closure we must have $-g \in \mathbb{Z}[x]$. Therefore, for any $\alpha \in \mathcal{A}$, we must have $-\alpha \in \mathcal{A}$.

I have absolutely no idea how you wanted us to show closure. Here's an approach I know of that works though.

Let $f, g \in \mathbb{Z}[x]$ be defined as

$$f(x) = \prod_{i=1}^n (x - \alpha_i), \quad g(x) = \prod_{j=1}^m (x - \beta_j),$$

respectively. In particular, let $\alpha_1 = \alpha$ and $\beta_1 = \beta$, that way α and β are roots for f and g , respectively. We can then define a polynomial

$$P(x) = \prod_{i=1}^n \prod_{j=1}^m (x - (\alpha_i + \beta_j)),$$

which clearly has a root of $(\alpha + \beta)$ in the term $i = j = 1$. Moreover, it is clearly still monic. Similarly, we can define

$$M(x) = \prod_{i=1}^n \prod_{j=1}^m (x - \alpha_i \beta_j),$$

which has a root of $\alpha\beta$ at $i = j = 1$ as well, and is also still clearly monic. Because of the intrinsic symmetry of the polynomials, you clearly have that $P, M \in \mathbb{Z}[x]$, and thus $\alpha\beta, \alpha + \beta \in \mathcal{A}$.

Since $\mathcal{A}$ is closed under addition, multiplication, negation, and is nonempty, we have that $\alpha - \beta \in \mathcal{A}$ and $\alpha\beta \in \mathcal{A}$ for any $\alpha, \beta \in \mathcal{A}$. By the subring test, $\mathcal{A}$ is a subring of $\mathbb{C}$.